AJ Barea

Ph.D. Student, Computing and Information Sciences · Rochester Institute of Technology

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Research Interests

Software engineering for federated and distributed machine learning systems; agentic AI with tool use and human oversight; testing, observability, and reproducibility infrastructure for ML research; trustworthy and privacy-preserving learning from human-generated data, including what shared model updates reveal about their source; empirical software engineering on code review; developer experience in AI research tooling.

Education

Doctor of Philosophy, Computing and Information Sciences Aug 2026 to present

Rochester Institute of Technology, Rochester, NY

- Advisor: Dr. Leon Reznik
- NSF AWARE-AI National Research Traineeship (NRT), AI Software track

Master of Science, Software Engineering Dec 2025

Rochester Institute of Technology, Rochester, NY

- GPA: 4.0 / 4.0
- Capstone: *Enhancing Federated Learning Execution Framework*
- Coursework: Software Architecture, Software Quality Assurance, Engineering Cloud Software Systems, Self-Adaptive Systems with Reinforcement Learning, Model-Driven Development, Software Construction, Collaborative Software Development

Bachelor of Science, Computer Engineering May 2022

University of South Florida, Tampa, FL

- Capstone: Private blockchain system for document certification (proof-of-work mining, SHA-256 document signatures, OAuth 2.0 / JWT admin dashboard)

Research Experience

Doctoral Researcher, Federated Agents Aug 2026 to present

Rochester Institute of Technology

- Leading a research direction on coding agents that learn each organization's conventions and pool what generalizes, while each organization decides what may leave
- Designed a pre-registered test of whether learned house style transfers at the project or the organization boundary, written up as a Stage 1 registered report in preparation for MSR 2027
- Built sphragis, an open study harness that collects public code review from Gerrit and GitHub under each host's terms, pseudonymizes it at ingestion, seals the test window by hash, and trains and scores LoRA adapters on Slurm GPU clusters
- Built a source-attribution measurement of whether the shared/private adapter splits in the federated personalization literature reveal the source organization in the half they transmit

Summer Research Scholar May 2026 to Aug 2026

Laboratory for Analytic Sciences (LAS), NC State University · SCADS (Summer Conference on Applied Data Science)

- Selected for a competitive summer research cohort applying data science to analytic sensemaking problems
- Prototyped an agentic sensemaking system on the Claude Agent SDK, tracing an entity's evidence across graph, relational, and text stores into a cited analytic note
- Built a governed MCP orchestration platform for agentic analytic workflows, cutting a measured policy-leak rate from 75% to 0% under a deterministic red-team evaluation battery
- Delivered a motion-first object tracking pipeline for camouflaged targets in video, benchmarked end to end against published state-of-the-art methods

- Instrumented agent and pipeline runs so multi-step reasoning could be inspected, replayed, and evaluated rather than treated as a black box

Graduate Research Assistant May 2025 to present

Laboratory of Data Quality and Intelligent Security (LDQIS), Rochester Institute of Technology · Advisor: Dr. Leon Reznik

- Designed a pytest suite reaching 80%+ coverage, run in GitHub Actions CI, for InteFL, the lab's distributed federated learning framework, converting a research prototype into a dependable, reproducible platform
- Built attack and defense tooling for InteFL: per-strategy schedules that start, stop, and stack label-flipping and Gaussian-noise poisoning by round, a multi-token sequence replacement attack, before/after snapshots of what each attack changed, and the FreqFed robust aggregation strategy
- Added Hugging Face dataset loaders for financial, legal, and biomedical text (including MEDAL) for transformer experiments, and let InteFL's experiment-planning AI agent run on Claude, any OpenAI-compatible vendor, or local Ollama and vLLM models
- Built the lab's new website (Astro, Tailwind, git-backed CMS) to replace its legacy Flask site

Publications

Refereed Journal

1. Korobeinikov, D., Zatsarenko, R., Chuprov, S., **Barea, A. J.**, & Reznik, L. (2026). InteFL Framework: Optimizing Federated Learning With Metacognition for Application Design and Deployment. *IEEE Intelligent Systems*, 41(4), 44–55. doi:10.1109/MIS.2026.3658072

Refereed Conference

2. Korobeinikov, D., **Barea, A. J.**, Reznik, L., Zatsarenko, R., Chuprov, S., & Peredo, A. (2026). InteFL: Framework for AI-Assisted Design of Trustworthy and Efficient Federated Learning Applications. *International Joint Conference on Neural Networks (IJCNN), IEEE World Congress on Computational Intelligence*, Maastricht, Netherlands, 1–6. doi:10.1109/IJCNN69854.2026.11730045

3. **Barea, A. J.**, Johnson, B. K., Tarosky, J. H., & Padilla, J. G. (2027). makesense: Governed Agentic Sensemaking over Heterogeneous Data Stores for Intelligence Analysis. *Innovative Applications of Artificial Intelligence (IAAI-27)*. Accepted as oral presentation.

Refereed Workshop

4. **Barea, A. J.** (2026). Kourai Khryseai: Transparent Human-on-the-Loop Multi-Agent Software Development. *North-East AI Agents Day 2026 Workshop*, Jane Street, New York, NY. Accepted as poster (sole author). OpenReview: jdD8sORLLY

Technical Report

5. Padilla, J. G., **Barea, A. J.**, Johnson, B. K., & Tarosky, J. H. (2026). makesense: A Governed Offline Interface for Multi-Source Intelligence Sensemaking. In *SCADS 2026 Technical Report*, 7–34. Laboratory for Analytic Sciences, NC State University. LAS release

6. Johnson, B. K., & **Barea, A. J.** (2026). video-motion: Motion-First Segmentation of Camouflaged Objects in Video at Field-Deployable Speed. In *SCADS 2026 Technical Report*, 253–284. Laboratory for Analytic Sciences, NC State University. LAS release

In Preparation

7. **Barea, A. J.** Do Organizations Leave a Learnable House Style in Code Review? Stage 1 registered report, *MSR 2027 Registered Reports* track with *Empirical Software Engineering*.

Software

8. **Barea, A. J.** (2026). techne: a Claude Code plugin marketplace for repositories, documents, and bibliographies. Zenodo. doi:10.5281/zenodo.23330765

9. **Barea, A. J.** (2026). keryx: gives Claude Code a voice. Zenodo. doi:10.5281/zenodo.23327640

Technical Skills

Machine Learning: PyTorch, Hugging Face Transformers and PEFT (LoRA), scikit-learn, Flower, federated learning, reinforcement learning, Webots, SAM 2, OpenCV

Agentic AI: Claude Agent SDK, Model Context Protocol (MCP), A2A protocol, tool-use orchestration, LLM evaluation, local models (Ollama, vLLM)

Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, Rust (PyO3, maturin)

Web and Services: FastAPI, React, Vue.js, Spring Boot, Node.js, Next.js, Astro, Electron

Infrastructure: Docker, Terraform, GitHub Actions, CI/CD, Slurm HPC clusters, AWS (Lambda, EC2, S3, DynamoDB, API Gateway, CloudFront, Rekognition, Comprehend, Translate)

Observability and QA: pytest, OpenTelemetry, Jaeger, Prometheus, coverage analysis, Playwright

Data Stores: PostgreSQL, Neo4j

Collaboration Tools: Git, Jira, Confluence

Selected Research Software

Ariadne · SCADS sensemaking prototype built on the Claude Agent SDK, with MCP tool servers, Neo4j and PostgreSQL knowledge stores, and OpenTelemetry tracing.

Kourai Khryseai · Multi-agent system using the A2A protocol and MCP with real-time speech synthesis, telemetry through OpenTelemetry, Jaeger, and Prometheus.

Velocity-FL · Federated learning research toolkit with a Rust core and Python bindings (maturin), PyTorch and Hugging Face model support, and an MCP server interface.

Phalanx-FL · Federated LoRA testbed on the Flower 1.39 app model with OpenTelemetry tracing that follows every round down to each client and records why each failed client failed.

techne · Claude Code plugin marketplace of agent workflows for repository maintenance, adversarial code review, LaTeX and slide builds, bibliography verification against arXiv and Crossref, Gmail and course-site reading, and idea triage against prior work; includes **keryx**, spoken session replies from a local model and local speech synthesis.

Pharos · Federated personalization testbed with security-labeled synthetic data, measuring what a personal/shared boundary costs and how much privacy leaks past it under secure aggregation and audit policy.

Security Evaluation System · Windows host security assessment that scores patches, open ports, firewall, antivirus, and password policy through a CLIPS expert system, a PyTorch network trained to approximate it, or federated training; FastAPI backend with a Next.js and Electron front end. Code

Orchestrate Triage · Support-ticket triage agent on the Claude API across three product domains, with corpus-grounded replies, structured tool-call output, prompt-injection defense, prompt caching, and the Message Batches API. Built solo for HackerRank Orchestrate 2026. Code

Quantum Federated Learning Privacy · Study of when variational quantum circuits keep a federated client's training data safe from gradient inversion, across encoding depth, trainable layers, batch size, training, and simulated IBM Heron device noise. CSCI-739 Quantum Machine Learning project, in progress. Code

Myo Lever Controller · Drives a simulated Clearpath Jackal from one forearm: the Myo armband's motion sensor sets speed and steering, and a per-driver EMG classifier reads a deliberate fist as the brake. Built at the AWARE-AI Fall Hackathon, where the team won Best Presentation. Code · Site

BioRadio EMG-to-MIDI · Real-time instrument: forearm EMG from the BioRadio streams over Lab Streaming Layer to a random-forest classifier of eight gestures, which plays chords and instruments through FluidSynth. Built at the AWARE-AI Spring Hackathon, where the team won Best AI Model.

Autonomous Robot Navigation (RL) · Tabular Q-learning agent in the Webots simulator with state discretization and reward shaping, plus a PyTorch DQN with experience replay and a target network. Code

Teaching Experience

Teaching Assistant, Software Architecture Aug 2025 to Dec 2025
Rochester Institute of Technology

- Mentored students in design patterns, architectural styles, and quality attributes and tactics
- Led recitation sections and office hours; taught a full class session during instructor absence
- Graded assignments with detailed feedback emphasizing rigorous engineering practice

Presentations and Posters

Talk, *Federated Learning for the Real World: Privacy, Security, and Cost, Without the Trade-offs* (Korobeinikov, D., **Barea, A. J.**, & Reznik, L.). 2026 Rochester Security Summit (RocISSA), Rochester, NY. October 2026. Session

Talk, *Federated Agents: Coding Agents That Learn Each Company's House Rules Without Its Code Leaving*. Rochester Institute of Technology. October 2026. Slides

Poster, *Kourai Khryseai: Transparent Human-on-the-Loop Multi-Agent Software Development*. North-East AI Agents Day 2026 Workshop, Jane Street, New York, NY. May 2026. OpenReview

Poster #26, *Narrative Interfaces for Multi-Agent Software Development*. CHAI Research Poster Session, RIT Center for Human-aware Artificial Intelligence, Student Innovation Hall, Rochester, NY. April 2026.

Talk, *TraceFL: Debugging Federated Learning Systems*. Software Quality Assurance, Rochester Institute of Technology. Fall 2025.

Tutorial series, applied ML, reinforcement learning, and AI developer tooling. YouTube (@ajbarea). 2024 to present.

Professional Experience

Software Engineer (Contract) Feb 2023 to Aug 2024

Lowe's Home Improvement, Remote · Distribution Management Systems

- Cut daily error logs by 150,000 entries across regional distribution centers through systematic debugging, log analysis, and performance work
- Built a full-stack internal tool for database querying and SQL script generation, reducing support ticket resolution time
- Authored standard operating procedures that reduced new engineer onboarding time by 80%
- Performed Tier 3 root cause analysis and delivered tested, documented production fixes

UX Developer (Intern) Mar 2022 to May 2022

Phiquest, Tampa, FL

- Designed wireframes for web applications with cross-functional teams
- Engineered a blockchain-based document authenticator Chrome extension in Node.js

Honors and Awards

NSF AWARE-AI National Research Traineeship, Rochester Institute of Technology	2026
SCADS Summer Research Scholar, Laboratory for Analytic Sciences (LAS), NC State University	2026
Best Presentation, AWARE-AI Fall Hackathon, Rochester Institute of Technology · team entry	2026
HackerRank Orchestrate 2026 Hackathon · 11th of 1,349 (top 0.8%), solo entry	2026
Best AI Model, AWARE-AI Spring Hackathon, Rochester Institute of Technology · team entry	2026
Certificate of Performance, Phiquest	2022
The National Society of Leadership and Success	2019

Service and Outreach

Science communication: Produce open educational content on AI development tooling, pair programming with AI assistants, and reinforcement learning for a public audience of students and developers (YouTube @ajbarea).

Open source: Maintain public research software and documentation across federated learning, agentic AI, and developer tooling projects (github.com/ajbarea).

Program participation: AWARE-AI Spring and Fall Hackathons (2026); AWARE-AI workshops on research storytelling and collaborative research; NSF SaTC-EDU AI4IA training on LLM-assisted cyber incident analysis.